Diseases and Treatment

I. Viral Diseases

Notes

- Word 'VIRUS' is originated from the Latin neuter 'virus' which means 'venomous substance'.
- The virus is an ultramicroscopic (20-300 nm in diameter), metabolically inert, infectious agent that replicates only within the cells of living hosts, mainly bacteria, plants and animals; composed of an RNA or DNA core, a protein coat and in more complex type, a surrounding envelope.
- The protein coat is known as **capsid** and its subunit is known as **capsomere**.
- Most viruses have either RNA or DNA as their genetic material, which may be single or double-stranded.
- The entire infectious virus particle is called **virion**. Virus is the nucleoprotein particle where as virion is the active, infectious form of the virus.
- It was discovered by Russian scientist **Dmitri Ivanovsky** in 1892. He found that a disease of tobacco plants could be transmitted by an agent, later called tobacco mosaic virus, passing through a minute filter that would not allow the passage of bacteria.
- In 1898, **Martinus Beijerinck** independently replicated Ivanovsky's filtration experiments and then showed that the infectious agent was able to reproduce and multiply in the host cells of the tobacco plant. He coined the term 'virus'.
- Tobacco mosaic virus was the first virus to be crystallized. It was achieved by **Wendel Meredith Stanley** in 1935 who also showed that TMV remains active after crystallization.
- The virus is a link between non-livings and livings. Viruses are non-livings when they are outside the host cell as they do not have any cellular machinery of their own. But when they are present inside the body of the host, they are living. They take over the host cell machinery to replicate themselves, eventually destroying the host cell.
- The river Ganga is self-cleansing and has healing powers, indeed its water has bacteriophages, who infect and kill bacteria.
- **Phage Therapy** is the use of bacteriophages to treat pathological infections caused by bacteria.
- The branch of biology which deals with the study of the virus is called **virology**.
- In 1977, India was declared to be free from smallpox.

Guinea Worm Disease (GWD) :

- Guinea worm disease (Dracunculiasis) was an important public health problem in many states of India before it was eradicated in 2000.
- It is a parasitic infection by the Guinea worm. It is caused by a large nematode, Dracunculus medinensis, which passes its life cycle in two hosts - Man and Cyclops (water fleas).
- A person becomes infected when they drink water that contains water fleas (Cyclops) infected with guinea worm larvae. Initially, there are no symptoms. About one year later, the female worm forms a painful blister in the skin, usually on the lower limb.

Polio :

- It is also called **poliomyelitis**.
- It is an infectious disease caused by the **poliovirus**.
- Poliovirus is usually spread from person to person through infected fecal matter entering the mouth. It may also be spread by food or water containing human feces and less commonly from infected saliva.
- Those who are infected may spread the disease for upto six weeks even if no symptoms are present.
- It may affect the spinal cord causing muscle weakness and paralysis because poliovirus destroys the cells of the brain and spinal cord-controlling the functions of muscles.
- The first polio vaccine was developed by **Jonas Salk** in 1952 and came into use in 1955.
- The oral polio vaccine was developed by **Albert Sabin** and came into commercial use in 1961.
- They are on the World Health Organization's List of Essential Medicines, the most effective and safe medicines needed in a health system.
- Trivalent Oral Polio Vaccine was used against for all three types of poliovirus (Type 01, Type 02 and Type 03).
- The World Health Organization (WHO) presented the official certification to India for its 'Polio Free' status on 27 March, 2014.
- In September, 2015 WHO declared that poliovirus type 2 has been eradicated from the earth- no cases caused by this serotype had been detected since November, 1999. For this reason WHO decided to remove the type 2 PV and switch from trivalent to bivalent vaccine in April, 2016.

Coronavirus Disease 2019 (COVID-19)

- Coronaviruses are a large family of RNA viruses. In humans, these viruses cause respiratory tract infections, ranging from the common cold to more severe diseases such as **SARS (Severe Acute Respiratory Syndrome;** first identified in 2002), **MERS (Middle East Respiratory Syndrome or camel flu**; first identified in 2012); and **COVID-19**.
- COVID-19 (Coronavirus Disease 2019) is an infectious disease caused by Novel Coronavirus (nCOV) or Severe Acute Resipiratory Syndrome Coronavirus 2 (SARS-CoV-2).
- It was first identified in December 2019 in **Wuhan** (China), and has resulted in an ongoing pandemic with millions of cases and lakhs of deaths worldwide.
- The World Health Organization (WHO) declared the COVID-19 outbreak a public health emergency of international concern on 30 January 2020 and a pandemic on 11 March 2020.
- Common signs of COVID-19 infection include respiratory symptoms, fever, cough, shortness of breath, breathing difficulties, fatigue and loss of smell and taste.
- While the majority of cases result in no or mild to moderate symptoms, some progress to acute respiratory distress syndrome (ARDS) likely precipitated by a cytokine storm, multi-organs failure, septic shock and severe blood clotting leading to death.
- Preventive measures include physical or social distancing, quarantining, ventilation of indoor spaces, covering coughs and sneezes, hand washing, and keeping unwashed hands away from the face. The use of face masks or coverings has been recommended in public settings to minimize the risk of transmissions.
- The primary treatment of this disease is currently symptomatic. Management involves the treatment of symptoms, supportive care, isolation and experimental measures.
- Several vaccines intended to provide aquired immunity against SARS-CoV-2, have been developed and various countries have initiated mass vaccination campaigns.
- Some prominent vaccines types authorized by at least one regulating authority for public use are : **RNA vaccines** (the **Pfizer-BioNtech vaccine** and **Moderna vaccine**), conventional inactivated SARS-CoV-2 vaccines (**BBIBP-CorV** from **Sinopharm**, **BBV152** or **Covaxin** from **Bharat Biotech**, **CoronaVac** from **Sinovac** and **WIBP** from **Sinopharm**), **viral vector** vaccines (**Sputnik V and Sputnik Light** from the Russia's **Gamaleya Research Institute**, the **Oxford - AstraZeneca vaccine** or **Covishield**, and **Ad5-nCoV** from **CanSino Biologics**), and **peptide vaccine** (**EpiVacCorona** from the **Vector Institute**).
- On 16 January 2021, India started its national vaccination programme against the SARS-CoV-2 with two vaccines : **Covishield** (Oxford - AstraZeneca vaccine manufactured by Pune-based Serum Institute of India) and **Covaxin** (Developed by Bharat Biotech in association with the Indian council of Medical Research and National Institute of Virology).
- Since then, India's drug regulator approved many other COVID-19 vaccines. Some of them are as follows : Sputnik V (manufactured under license by Dr. Reddy's Laboratories, with additional production from Serum Institute of India being started in September, 2021), Moderna vaccine, Johnson & Johnson vaccine, ZyCoV-D (world's first DNA based COVID-19 vaccine; manufactured by Zydus Cadila, India), Corbevax (developed by Texas Children's Hospital Center for Vaccine Development and Baylor College of Medicine in Houston, Texas and Dynavax technologies, USA and licensed to Indian biopharmaceutical firm Biological E. Limited), Covovax (developed by Novavax and the Coalition for Epidemic Preparedness Innovations-CEPI) and Sputnik Light etc.
- The countrywide vaccination drive in India was rolled out on 16 January, 2021 with healthcare workers (HCWs) getting inoculated in the first phase. The vaccination of frontline workers (FLWs) started from 2 February, 2021.
- The next phase of COVID-19 vaccination commenced from 1 March, 2021 for people over 60 years of age and those aged 45 and above with specified co-morbid conditions. The country launched vaccination for all aged more than 45 years from 1 April, 2021. The government then decided to expand its vaccination drive by allowing everyone above 18 to be vaccinated from 1 May, 2021.

- The next phase of COVID-19 vaccination commenced from 3 January, 2022 for adolescents in the age group of 15-18 years.
- India began administering precaution booster doses of COVID-19 vaccine to healthcare workers, frontline workers, including personnel deployed for election duty and those aged 60 and above with co-morbidities, from 10 January, 2022 amid a spike in coronavirus infections fuelled by Omicron variant of the virus in the country.
- COVID-19 vaccination of children in the age group of 12 to 14 years started from 16 March, 2022 with only Corbevax vaccine.
- The administration of precaution booster dose of COVID-19 vaccine to 18-plus population through private vaccination centres started from 10 April, 2022.
- As on 19 June, 2024, India has administered over 2.2 billion doses overall, including first, second and Precautionary (booster) doses of the approved vaccines.
- As on 24 March, 2025, total official deathcount due to COVID-19 in India is 533664.

Rhinovirus :

- The Rhinovirus (from Greek rhinos 'of the nose' and the Latin 'virus') is the most common viral infectious agent in humans and is the predominant cause of the common cold.
- Rhinovirus infection proliferates in temperatures of 33-35ºC (91-95ºF), the temperatures found in the nose.
- The three species of rhinovirus (A, B and C) include around 160 recognized types of human rhinoviruses that differ according to their surface proteins (serotypes).
- Its symptoms include sore throat, runny nose, nasal congestion, sneezing and cough, fatigue, headache etc.

Hepatitis-B :

- Hepatitis-B is an infectious disease caused by the hepatitis-B virus (HBV) that affects the liver.
- It can cause both acute and chronic infections. It can cause scarring of the organ, liver cirrhosis and cancer.
- It spreads when people come in contact with the blood, open sores, or body fluids of someone who has the hepatitis-B virus. Infection around the time of birth or form contact with other people's blood during childhood is the most frequent method by which hepatitis-B is acquired.
- Hepatitis-B symptoms include jaundice, fever, fatigue that persists for weeks or months, stomach trouble like loss of appetite, nausea and vomiting and belly pain.

Mumps :

- Mumps is a contagious disease caused by a virus (mumps virus) that passes from one person to another through saliva, nasal secretions, and close personal contact.
- The condition primarily affects the salivary glands, also called the parotid gland.
- The hallmark symptom of mumps is swelling of the salivary glands. Initial signs and symptoms often include fever, muscle pain, headache, poor appetite, and feeling generally unwell.
- Symptoms are often more severe in adults than in children. Complications may include meningitis, pancreatitis, permanent deafness and testicular inflammation, which uncommonly results in infertility.

Rabies (Hydrophobia) :

- Rabies lyssavirus, formerly rabies virus, is a neurotropic virus that causes rabies in humans and animals.
- Rabies transmission can occur through the saliva of animals like dog, cat, bat and wild animals like fox, etc. It is spread when an infected animal scratches or bites another animal or human.
- Early symptoms can include fever and tingling at the site of exposure. These symtoms are followed by one or more of the following symptoms : violent movements, uncontrolled excitement, fear of water, confusion, and loss of consciousness. Once symptoms appear, the result is nearly always death.

Herpes Disease :

- It is a viral infectious disease. It is caused by the herpes simplex virus.
- It causes sores around the mouth and lips and genital organs.

Meningitis :

- Meningitis is an inflammation of the meninges. The meninges are the three membranes that cover the brain and spinal cord. Meningitis can occur when fluid surrounding the meninges becomes infected.
- The most common symptoms are fever, headache and neck stiffness. Other symptoms include confusion or altered consciousness, vomiting and an inability to tolerate light or loud noises.
- Meningitis may be caused by infection with viruses, bacteria, or other microorganisms. Meningitis caused by **meningococcal** bacteria may be accompanied by a characteristic rash.

Dengue Fever :

- Dengue fever is transmitted by the bite of an Aedes mosquito (mainly A. aegypti) infected with a dengue virus.Tiger mosquito (Aedes albopictus) can also serve as a vector for this virus.
- Symptoms typically begin three to fourteen days after infection.

- This may include a high fever, headache, vomiting, muscle and joint pains and a characteristic skin rash.
- It is also known as breakbone fever.
- Dengue fever affects the number of platelets in the blood. Dengue virus, the main cause of dengue fever induces bone marrow suppression. Since bone marrow is the manufacturing centre of blood cells its suppression causes deficiency of blood cells leading to low platelet count.

Chikungunya :

- It is a viral disease transmitted to humans by chikungunya virus (CHIKV) infected mosquitos Aedes albopictus (the Tiger mosquito) and Aedes aegypti.
- It causes fever and severe joint pain. Other symptoms include muscle pain, headache, nausea, fatigue and rash.
- It is not a contagious disease because an infected man cannot spread the infection directly to other persons.

Zika Fever :

- It is a viral infectious disease caused by the Zika Virus.
- Symptoms may include fever, red eyes, joint pain, headache and maculopapular rash.
- Zika virus is transmitted to humans primarily through the bite of infected Aedes aegypti and Aedes albopictus.
- It can also be sexually transmitted and potentially spread by blood transfusions. Infections in pregnant women can spread to the baby.

Yellow Fever :

- Yellow fever is an acute viral haemorrhagic disease transmitted by yellow fever virus infected mosquito (mainly Aedes aegypti). Tiger mosquito (Aedes albopicturs) can also serve as a vector for this virus.
- Symptoms of yellow fever include fever, headache, jaundice, muscle pain, nausea, vomiting and fatigue.
- Yellow fever is prevented by an extremely effective vaccine, which is safe and affordable.

Japanese Encephalitis (JE) :

- Japanese Encephalitis (JE) is an infection of the brain caused by the Japanese encephalitis virus (JEV).
- While most infections result in little or no symptoms, occasional inflammation of the brain occurs. In these cases, symptoms may include headache, vomiting, fever, confusion and seizures.
- There is no specific treatment or cure for JE. Once a person has the disease, treatment can only relieve the symptoms. Prevention and care are the best form of treatment of JE.
- JE is a mosquito-borne viral infection. The species of Culex mosquitoes are the main vector of JE.

Acquired Human Immunodeficiency Syndrome (AIDS) :

- AIDS is a spectrum of conditions caused by infection with the **human immunodeficiency virus (HIV)**. It is a fatal disease.
- The virus responsible for AIDS belongs to the retro group of viruses.
- It destroys the immune system, the body's defence against infections, leaving an individual vulnerable to a variety of other infections and certain malignancies that eventually cause death.
- HIV is not spread by coughing, sneezing or casual contact (e.g. shaking hands).
- HIV is fragile and cannot survive long outside the body, therefore a direct transfer of bodily fluids is required for transmission as - unsafe sexual contact, contaminated blood transfusion and to a child from infected mother with the contact of the placenta.
- **Western blot** and **ELISA** (enzyme-linked immunosorbent assay) tests are used to detect the AIDS. Western Blot is used to confirm a positive ELISA and the combined tests are 99.9% accurate.
- **World AIDS Day –** designated on 1 Dec. every year since 1988, is an international Day dedicated to raising awareness of the AIDS pandemic caused by the spread of HIV infection and mourning those who have died of the disease.

Ebola Virus Disease :

- Ebola virus disease (Ebola haemorrhagic fever) is a rare and deadly disease in people and non-human primates.
- The disease was first identified in 1976 in two simultaneous outbreaks : one in Nzara (South Sudan) and the other in Yambuku (DR Congo), a village near the **Ebola river** from which the disease takes its name.
- The virus that causes EVD are located mainly in, Sub-Saharan Africa. People can get EVD virus through direct contact with an animal (bat and non-human primates) or a sick or dead person infected with Ebola virus.
- It causes fever, body aches and diarrhoea and sometimes bleeding inside and outside the body. As the virus spreads through the body, it damages the immune system and organs ultimately and this leads to severe uncontrollable bleeding. The disease has a high risk of death, killing between 25 to 90 percent of those infected.
- The largest Ebola outbreak to date was the epidemic in West Africa (Dec. 2013 – Jan. 2016) with 28,646 cases and 11,323 deaths.

Nipah Virus Infection :

- A Nipah virus infection is a viral infection caused by the Nipah virus (Ni V). In May 2018, an outbreak of this disease resulted in 17 deaths in Kerala (India).
- In September 2021, Nipah virus resurfaced in Kerala, claiming the life of a 12 year old boy.
- In July-September 2024, an outbreak was confirmed again in Kerala, resulting in 2 deaths.

- This disease was first identified in Malaysia in 1998, and it is named after a village in Malaysia, **Sungai Nipah.**
- Symptoms from infection vary from none to fever, cough, headache, shortness of breath, and confusion. This may worsen into a coma over a day or two.
- Complications can include inflammation of the brain and seizures following recovery.
- The Nipah virus (NiV) is a type of RNA virus. It normally circulates among specific types of fruit bats.
- Its management involves supportive care. There is no vaccine or specific treatment till date. Prevention is by avoiding exposure to bats and sick pigs and not drinking raw date palm sap (palm toddy).

- **Endemic –** (of a disease or condition) regularly found among particular people or in a certain area.
- **Epidemic –** The rapid spread of a particular disease to a large number of people in a given population within a short period of time.
- **Pandemic –** (of an infectious disease) that has spread across a large region, for instance whole country, multiple continents or worldwide, affecting a substantial number of people.

Question Bank

1. The disease causing microorganism among these is :

(a) Archaea (b) Pathogens
(c) Bacteria (d) Protozoa

U.P.P.C.S. (Pre) 2024

Ans. (b)

A pathogen is a living microorganism that can cause disease in another organism, such as a human. Typically, the term pathogen is used to describe an infectious microorganism, such as a virus, bacterium, protozoan, prion, viroid, or fungus. They can also be referred to as infectious agents or germs. Pathogens can spread through the air, by direct or indirect contact, or through blood, breast milk, or other body fluids. It is to be noted that while many bacteria and protozoa are pathogens, many of them are harmless or even beneficial.

2. What is not true about virus?

(a) Virus term was coined by Dutch microbiologist M.W. Beijerinck.
(b) Nucleic acid genome of virus can be DNA or RNA-based.
(c) DNA-based viruses are usually single-stranded.
(d) RNA-based viruses accumulate more genetic changes (mutations) as compared to DNA-based viruses.

Chhattisgarh P.C.S. (Pre) 2021

Ans. (c)

A DNA virus is a virus that has a genome made of deoxyribonucleic acid (DNA) that is replicated by a DNA polymerase. They can be divided between those that have two strands of DNA in their genome, called double-stranded DNA (dsDNA) viruses, and those that have one strand of DNA in their genome, called single-stranded DNA (ssDNA) viruses. Thus, DNA-based viruses are not usually single-stranded. Statements of other three options are correct.

3. A bacteriophage is a :

(a) bacterium with a tail
(b) newly formed bacterium
(c) bacterium infecting virus
(d) virus infecting bacterium

M.P.P.C.S. (Pre) 2017

Ans. (d)

A virus infecting bacterium is called bacteriophage. It infects and replicated within bacteria and archaea. Like other viruses, its outer covering is made up of protein and nucleic acid (DNA or RNA) is present in it.

4. Which of the following statements is/are correct?

1. Viruses lack enzymes necessary for the generation of energy.
2. Viruses can be cultured in any synthetic medium.
3. Viruses are transmitted from one organism to another by biological vectors only.

Select the correct answer using the codes given below:

(a) Only 1 (b) Only 2
(c) 1 and 3 (d) All of these

I.A.S. (Pre) 2013

Ans. (a)

A virus is a small infectious agent that replicates only inside the living cell of other organisms. Energy producing system is absent in them as viruses lack enzymes necessary for the generation of energy. Thus, they cannot be cultured in any synthetic medium. Viruses exist in the form of independent particles. The viral particles also known as virions. Viruses are transmitted from one organism to another by various means e.g. by air, water, food, insects, contact etc.

5. Which of the following does not have any enzyme in its cells?

(a) Lichen (b) Virus
(c) Bacteria (d) Algae
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (e)

Lichen, bacteria and algae, all these have enzymes in its cells. The virus generally relies on the enzymes already present in the host cell or make enzymes that it needs using its own genome inside the host cell. Some viruses have no enzymes at all inside the viral particle itself. In other viruses, a small number of enzymes can be found inside the viral particle itself and also on the surface of some viruses. Enzymes found in some viral particles are reverse transcriptase, RNA-dependent RNA polymerase, an intigrase, neuraminidase etc. Thus, the option (e) is the right answer.

6. The term 'ACE2' is talked about in the context of :

(a) genes introduced in the genetically modified plants
(b) development of India's own satellite navigation system
(c) radio collars for wildlife tracking
(d) spread of viral diseases

I.A.S. (Pre) 2021

Ans. (d)

Angiotensin-converting enzyme 2 (ACE2) is a protein on the surface of many cell types. It is an enzyme that generates small proteins – by cutting up the larger protein angiotensinogen – that then go on to regulate functions in the cell. ACE2 is present in many cells types and tissues including the lungs, heart, blood, vessels, kidneys, liver and gastrointestinal tract. It is present in epithelial cells, which line certain tissues and create protective barriers. ACE2 is a vital element in a biochemical pathway that is critical to regulating processes such as blood pressure, wound healing and inflammation. ACE2 also serves as the entry point into cells for some coronaviruses, including HCoV-NL63, SARS-CoV, and SARS-CoV-2. The SARS-CoV-2 spike protein itself is known to damage the endothelium via downregulation of ACE2. Using the spike-like protein on its surface, the SARS-CoV-2 virus binds to ACE2 – like a key being inserted into a lock – prior to entry and infection of cells. Hence, ACE2 acts as a cellular doorway – a receptor – for the virus that causes COVID-19.

7. Which of the following disease is related to Coronavirus?

(a) MERS (b) SARS
(c) COVID-19 (d) All of the above

M.P. P.C.S. (Pre) 2020

Ans. (d)

Coronaviruses are a group of related RNA viruses that cause diseases in mammals and birds. In humans and birds, they cause respiratory tract infections that can range from mild to lethal. Mild illnesses in humans include some cases of the common cold (which is also caused by other viruses, predominantly rhinoviruses), while more lethal varieties can cause SARS (Severe Acute Respiratory Syndrome), MERS (Middle East Respiratory Syndrome) and COVID-19 (Coronavirus Disease 2019), which is causing an ongoing pandemic. In cows and pigs they cause diarrhea, while in mice they cause hepatitis and encephalomyelitis.

8. The recent global pause during COVID-19 pandemic was caused due to :

(a) SARS-CoV-1 (b) H5N1
(c) SARS-CoV-2 (d) MERS-CoV-2

U.P.P.C.S. (Pre) 2024

Ans. (c)

The COVID-19 pandemic (also known as the coronavirus pandemic and COVID pandemic), caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), begain with an outbreak of COVID-19 in Wuhan, China, in December 2019. Soon after, it spread to other areas of Asia, and then worldwide in early 2020. The World Health Organization (WHO) declared the outbreak a public health emergency of international concern (PHEIC) on 30 January, 2020, and assessed the outbreak as having become a pandemic on 11 March, 2020.

9. In the context of vaccines manufactured to prevent COVID-19 pandemic, consider the following statements:

1. The Serum Institute of India produced COVID-19 vaccine named Covishield using mRNA platform.
2. Sputnik V vaccine is manufactured using vector based platform.
3. Covaxin is an inactivated pathogen based vaccine.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2022

Ans. (b)

For vaccination against COVID-19, India initially approved the Oxford-AstraZeneca vaccine (manufactured in India under license by Pune-based Serum Institute of India under the trade name Covishield) and Covaxin (a vaccine developed by Bharat Biotech in association with the Indian Council of Medical Research and National Institute of Virology). They have since been joined by the Russian Sputnik V (manufactured under license by Dr. Reddy's Laboratories, with additional production from Serum Institute of India) and some other vaccines. Covishield and Sputnik V both are examples of non-replicating viral vector vaccines, using an adenovirus shell containing DNA that encodes a SARS-CoV-2 protein while Covaxin is a whole virion inactivated SARS-CoV-2 based vaccine. Pfizer-BioNtech and Moderna vaccines, have been developed by using mRNA platform. Hence statement 1 is incorrect while statements 2 and 3 are correct.

10. Covishield, the COVID vaccine of India which is approved by WHO, is manufactured by :

(a) Serum Institute (b) Bharat Biotech
(c) Panacea Biotec (d) Zydus Cadila

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (a)

See the explanation of above question.

11. What is the volume of each dose of Covishield Vaccine given to adults?

(a) 1 ml (b) 2 ml
(c) 0.5 ml (d) 1.5 ml

Chhattisgarh P.C.S. (Pre) 2021

Ans. (c)

Covishield Vaccine has been developed by AstraZeneca with Oxford University in the UK and is being manufactured by the Serum Institute of India (SII) in Pune. Covishield Vaccine contains a non-replicating virus (a weakened chimpanzee adenovirus that causes common cold), that is genetically engineered to produce coronavirus proteins in the body, but the virus is weakened and cannot cause the disease. It provides active immunization against COVID-19 infection. Covishield regimen consists of two doses of 0.5 ml each. The Indian government has recommended that the time interval between 1st and 2nd doses should be between 12-16 weeks.

12. Which country exported Sputnik V COVID vaccine to India?

(a) America (b) Russia
(c) UK (d) France
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (b)

Sputnik V or Gam-COVID-Vac is an adenovirus viral vector vaccine for COVID-19 developed by the Gameleya Research Institute of Epidemiology and Microbiology in Russia. It is the world's first registered combination vector vaccine for the prevention of COVID-19, having been registered on 11 August, 2020 by the Russian Ministry of Health. Russia exported Sputnik V COVID vaccine to India.

13. Which of the following is mRNA vaccine used against Covid-19 infection?

(a) Sinovac (b) Novavax
(c) Moderna (d) Sputnik

U.P. P.C.S. (Pre) 2023

Ans. (c)

The Moderna COVID-19 vaccine is developed by the American company Moderna, the United States National Institute of Allergy and Infectious Diseases (NIAID), and the Biomedical Advanced Research and Development Authority (BARDA). It is an mRNA vaccine composed of nucleoside modified mRNA (modRNA) encoding a spike protein of SARS-CoV-2, which is encapsulated in lipid nanoparticles.
Sputnik-V – Adenovirus viral vector COVID-19 vaccine (Russia).
Novavax – Protein-based recombinant nanoparticle COVID-19 vaccine (USA).
Sinovac – Whole inactivated virus COVID-19 vaccine (China).

14. How do COVID vaccines stimulate an immune response?

(a) By introducing live attenuated SARS-CoV-2 virus
(b) By introducing a real SARS-CoV-2 virus
(c) By introducing a harmless piece of SARS-CoV-2 virus
(d) By introducing antibodies against SARS-CoV-2 virus

69th B.P.S.C. (Pre) 2023

Ans. (a)

Immune responses can be stimulated in host by introducing live, attenuated pathogen i.e. SARS-CoV2 virus. Option (b) and option (d) are not associated with the mechanism of stimulating immune response by covid vaccines. The statement made in option (c) is unscientific. Introduction of adjuvanated spike protein can't be considered as harmless piece, since it is the main component of the virus through which it enters the host and spread the infection. Hence, option (c) is ruled out and option (a) is the correct answer.

15. How do vector vaccines work to provide immunity?

(a) By introducing a weakened or inactivated virus into the body
(b) By directly attacking and destroying pathogens in the body
(c) By placing the virus in a modified version of a different virus
(d) By entering directly into the cells and enabling them to create spike proteins

69th B.P.S.C. (Pre) 2023

Ans. (c)

Viral vector vaccines use a modified version of a different virus as a vector to deliver protection. Viral vector vaccines use a harmless virus to smuggle the instructions for making antigens from the disease-causing virus into cells, triggering protective immunity against it.

16. 'Wolbachia method' is sometimes talked about with reference to which one of the following?

(a) Controlling the viral diseases spread by mosquitoes
(b) Converting crop residues into packing material
(c) Producing biodegradable plastics
(d) Producing biochar from thermo-chemical conversion of biomass

I.A.S. (Pre) 2023

Ans. (a)

Wolbachia is one of the world's most common types of bacteria, present in 50-60% of all insect species, including bees, beetles, butterflies, moths, and fruit flies. Wolbachia blocks viruses like dengue, chikungunya and Zika from growing in the bodies of Aedes aegypti mosquitoes. This means that Wolbachia-infected mosquitoes have a reduced ability to transmit viruses to people. When Wolbachia is established in a mosquito population it results in a decreasing incidence of dengue, Zika, chikungunya etc. World Mosquito Program's Wolbachia method can protect communities from mosquito-borne diseases without posing risk to natural ecosystems.

17. Which disease among these is transmitted by the bite of an insect?

(a) Scurvy (b) Dengue
(c) Pneumonia (d) Asthma

R.A.S./R.T.S. (Pre) 1996

Ans. (b)

Dengue fever is transmitted by the bite of an Aedes mosquito (mainly A. aegypti) infected with a dengue virus. It is also known as breakbone fever.

18. The dengue fever is caused by the bite of which of the following mosquitoes?

(a) Aedes (b) Anopheles
(c) Asian tiger mosquito (d) Culex

U.P. P.C.S. (Pre) 2022

Ans. (a)

See the explanation of above question.

19. Dengue is a fever caused and transmitted to another human by :

(a) Virus and female Aedes mosquito
(b) Bacteria and female Culex mosquito
(c) Fungus and female Aedes mosquito
(d) Protozoan and female Anopheles mosquito

M.P.P.C.S. (Pre) 2012

Ans. (a)

See the explanation of above question.

20. Which of these decreases in human the body due to dengue fever ?

(a) Platelets (b) Hb
(c) Sugar (d) Water

U.P.P.C.S. (Pre) 2012

Ans. (a)

Platelets are rapidly decreased due to dengue fever. These are important to prevent bleeding in the body. So due to decreasing level of platelets the patient may suffer from internal bleeding.

21. Consider the following statements :

1. Dengue is a protozoan disease transmitted by mosquitoes.
2. Retro-orbital pain is not a symptom of dengue.
3. Skin rash and bleeding from nose and gums are some of the symptoms of dengue.

Which of the statements given above is/are correct ?

(a) 1 and 2 (b) 3 only
(c) 2 only (d) 1 and 3

I.A.S. (Pre) 2005

Ans. (b)

Dengue is a viral disease transmitted by mosquitoes. Dengue fever is a painful, debilitating disease caused by viruses. Symptoms may include sudden high fever, severe headaches, pain behind the eyes (retro-orbital pain), severe joint and muscle pain, nausea, vomiting, skin rash and bleeding from nose and gums. These symptoms appear three to four days after the fever. Hence, statements 1 and 2 are incorrect, while statement 3 is correct.

22. Which one of the following diseases is not a bacterial disease?

(a) Diphtheria (b) Plague
(c) Pneumonia (d) Dengue

Rajasthan P.C.S. (Pre) 2023

Ans. (d)

Dengue is a viral disease transmitted by the bite of an Aedes mosquito infected with a dengue virus. Diphtheria, plague and pneumonia are bacterial diseases. Among these pneumonia is an infection of the lungs caused by bacteria, fungi, or viruses.

23. Yellow fever is spread by –

(a) Air (b) Water
(c) Housefly (d) None of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (d)

Yellow fever is contagious viral disease. In most cases, symptoms include fever, chills, loss of appetite, nausea, muscle pain particularly in the back and headaches. The disease is caused by the yellow fever virus and is spread by the bite of the female mosquito 'Aedes aegypti'.

24. Which one of the following diseases is not transmitted by tiger mosquitoes?

(a) Yellow fever (b) Dengue
(c) Chikungunya (d) Japanese Encephalitis

U.P.P.C.S. (Pre) 2013

Ans. (d)

Aedes albopictus also known as tiger mosquito or 'forest mosquito' is native to the tropical and subtropical areas of Southeast Asia. Aedes albopictus is an epidemiologically important factor for the transmission of many viral pathogens, including the yellow fever virus, dengue virus, and chikungunya virus, etc. Japanese encephalitis is transmitted by Culex species of mosquito.

25. Hydrophobia is caused by –

(a) Bacteria (b) Fungus

(c) Virus (d) Protozoan

U.P.P.C.S. (Mains) 2008

Ans. (c)

Fear of water is known as hydrophobia. Such type of patients gets afraid from river, lake, sea etc. Hydrophobia is the old name of Rabies which is a viral disease. Its virus affects the nervous system.

26. The disease caused by swelling of the membrane over spinal cord and brain is

(a) Leukaemia (b) Paralysis

(c) Sclerosis (d) Meningitis

U.P.P.C.S. (Pre) 2008

Ans. (d)

Meningitis is a disease caused by the inflammation of the protective membranes covering the brain and spinal cord known as meninges. Meningitis may be caused by infection with viruses, bacteria or other microorganisms.

27. One out of every 200 babies born in India, dies of diarrhoea caused by :

(a) Bacteria (b) Rotavirus

(c) Amoeba (d) Fungus

U.P.P.C.S. (Mains) 2010

U.P.P.S.C. (GIC) 2010

Ans. (b)

The rotavirus is a group of RNA viruses, some of which cause acute enteritis in humans. Rotavirus is the most common cause of severe diarrhoea among infants and young children. It is a genus of double-stranded RNA virus in the family Reoviridae.

28. If a human disease breaks out across a large region of the world, what is it called?

(a) Pandemic (b) Epidemic

(c) Endemic (d) Epizootic

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (a)

Endemic – A disease regularly found in a particular place or among a particular group of a people and difficult to get rid of.

Epidemic – A large number of cases of a particular disease happening in a given population within a short period of time.

Pandemic – An infectious disease that has spread across a large region, for instance whole country, multiple continents or worldwide, affecting a substantial number of people.

Epizootic – A disease that affects a large number of animals in some particular region within a short period of time.

29. Of the following which set includes all viral diseases ?

(a) Tuberculosis, Herpes, Rabies

(b) Mumps, Rabies, Herpes

(c) Cancer, Tuberculosis, Poliomyelitis

(d) Chicken Pox, Cancer, Tuberculosis

Uttarakhand P.C.S. (Mains) 2002

Ans. (b)

Common viral diseases include :
Chicken pox
Rabies
Flu (influenza)
Human immunodeficiency virus (HIV/AIDS)
Mumps, measles, poliomyelitis etc.

30. The poliovirus enters into the body through :

(a) Dog bite (b) Mosquito bite

(c) Polluted food and water (d) Saliva

Uttarakhand P.C.S. (Pre) 2003

U.P.P.C.S. (Pre) 1993

Ans. (c)

Polio is a highly contagious viral infection that can lead to paralysis, breathing problems or even death. The term poliomyelitis is from the Greek poliós meaning grey, myelós meaning the spinal cord. Poliomyelitis is highly contagious via the fecal-oral (intestinal source) and the oral-oral (oropharyngeal source) routes. In endemic areas, wild polioviruses can infect virtually the entire human population. The poliovirus enters into the body through polluted food and water.

31. 'Polio' is caused by :

(a) Bacteria (b) Virus

(c) Insects or Flies (c) Fungi

U.P.P.C.S. (Mains) 2007

Ans. (b)

See the explanation of above question.

32. Who discovered the polio vaccine ?

(a) Alexander Flemming (b) Jones Salk

(c) Robert Koach (d) Edward Genere

U.P.P.C.S. (Pre) 1991

Ans. (b)

Polio vaccine was discovered by Jones Salk in 1952. It is a dangerous disease in children which makes them handicapped. It is caused from poliovirus which affects central nervous system (CNS).

33. The vaccine for polio was first prepared by :

(a) Paul Ehrlich (b) Jones Salk

(c) Louis Pasteur (d) Joseph Lister

U.P.P.C.S. (Pre) 1995

Ans. (b)

See the explanation of above question.

34. Salk's vaccine is connected with which one of the following diseases?

(a) Smallpox (b) Tetanus

(c) T.B. (d) Polio

U.P.P.C.S (Pre) 2010

Ans. (d)

See the explanation of above question.

35. Against which of the following diseases has Government of India decided to given Bivalent ORV in place of Trivalent?

(a) Diptheria (b) Malaria

(c) Typhoid (d) Polio

R.A.S./R.T.S. (Pre) 2016

Ans. (d)

With India remaining polio-free for five years, the Government of India decided to switch to bivalent oral polio vaccine (OPV) in place of trivalent version. Trivalent OPV contains live and weakened versions for all the three types (1, 2 and 3) of wild poliovirus while the bivalent vaccine will contain type 1 and 3.

36. The pathogen responsible for the common cold is :

(a) Orthomixo virus (b) Rhinovirus

(c) Leukaemia virus (d) Poliovirus

U.P.P.C.S. (Pre) 1996

Ans. (b)

Rhinoviruses are the most common viral infectious agents in humans and are the predominant cause of the common cold. It may also cause some sore throats, ear infection, sinus infection and to a lesser degree, pneumonia and bronchiolitis.

37. Pneumonia is an infection of :

(a) nerve (b) blood

(c) skin (d) lungs

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (d)

Pneumonia is an infection of lungs. Pneumonia is an infection that inflames the air sacs in one or both lungs. The air sacs may fill with fluid or pus (purulent material), causing cough with phlegm or pus, fever, chills, and difficulty in breathing. A variety of organisms, including viruses, bacteria and fungi, can cause pneumonia.

38. Which one of the following diseases is not due to contamination of water ?

(a) Typhoid (b) Hepatitis B

(c) Jaundice (d) Cholera

R.A.S./R.T.S. (Pre) (Re. Exam) 2013

Ans. (b)

The hepatitis B virus (HBV) is known as a blood-borne virus because it is transmitted from one person to another via blood or fluids contaminated with blood. Another important way of transmission is from an infected mother to a newborn child, which occurs during or shortly after birth. It is not caused due to contamination of water. Typhoid, jaundice and cholera are waterborne diseases.

39. The liver disease Hepatitis-B is caused by –

(a) DNA Virus (b) RNA Virus

(c) Bacterium (d) Platyhelminth

U.P.P.C.S. (Mains) 2011

Ans. (a)

Hepatitis-B virus, abbreviated HBV, is a partially double-stranded DNA virus, a species of the genus Othohepadnavirus and a member of the Hepadnaviridae family of viruses. It causes scarring of the liver, liver failure, liver cancer and even death.

40. Hepatitis-B is caused by which micro-organism?

(a) Virus (b) Protozoa

(c) Bacteria (d) None of these

Jharkhand P.C.S. (Pre) 2010

Ans. (a)

See the explanation of above question.

41. Which one of the following statements is not correct?

(a) Hepatitis B virus is transmitted much like HIV.

(b) Hepatitis B, unlike Hepatitis C, does not have a vaccine.

(c) Globally, the numbers of people infected with Hepatitis B and C viruses are several times more than those infected with HIV.

(d) Some of those infected with Hepatitis B and C viruses do not show the symptoms for many years.

I.A.S. (Pre) 2019

Ans. (b)

Hepatitis is inflammation of the liver tissue which is most commonly caused by the Hepatitis viruses (A, B, C, D and E). Hepatitis B virus transmitted through receipt of contaminated blood or blood products, from mother to baby at birth, and also by sexual contact. Hence, Hepatitis B virus is transmitted much like HIV. According to WHO estimates globally, in 2022 the number of people infected with Hepatitis B virus is about 254 million and with Hepatitis C virus is about 50 million. On the other hand, till 2023 about 88.4 million (71.3 – 112.8 million) people have been infected overall with the HIV in which about 42.3 million (35.7 – 51.1 million) have died. Globally, 39.9 million (36.1 – 44.6 million) people were living with HIV at the end of 2023. Some of those infected with Hepatitis B and C viruses do not show the symptoms for many years. Thus, statements of option (a), (c) and (d) are correct.

Statement of option (b) is not correct because Hepatitis B has a vaccine that is recommended for all infants at birth and for children up to 18 years. Hepatitis C, unlike Hepatitis B, does not have a vaccine.

42. The organ of the human body directly affected by the disease of hepatitis is –

(a) Liver (b) Lungs
(c) Heart (d) Brain

U.P. Lower Sub. (Spl.) (Pre) 2004

Ans. (a)

See the explanation of above question.

43. Which of the following disease is caused by a virus?

(a) Diphtheria (b) Malaria
(c) Cholera (d) Hepatitis

U.P.U.D.A./L.D.A. (Pre) 2006

Ans. (d)

See the explanation of above question.

44. Which of the following pairs is incorrect?

(a) Plague-Rat (b) Rabies-Dog
(c) Tapeworm-Pig (d) Polio-Monkey

M.P.P.C.S. (Pre) 1999

Ans. (d)

Plague is an infectious disease which is spread by rats. Rabies or hydrophobia is caused by rabies virus. The virus is spread by the saliva of infected animals such as the bite of a mad dog. Tapeworm is caused by eating raw or undercooked pork whereas polio enters in human body by poliovirus which is contagious and usually spread from person to person through infected fecal matter entering the mouth.

45. Consider the following diseases :
1. Diphtheria, 2. Chickenpox, 3. Smallpox
Which of the above diseases has/have been eradicated in India?

(a) 1 and 2 only (b) 3 only
(c) 1, 2 and 3 (d) None

I.A.S. (Pre) 2014

Ans. (b)

India was declared smallpox free in 1977. Guinea worm disease (Dracunculiasis) was a major public health problem in many States of India before it was declared eradicated in 2000. On 27th March, 2014 the World Health Organisation (WHO) presented official certification to India for its 'Polio Free' status. In 2015, India was declared free from maternal and neonatal tetanus. Diphtheria and chickenpox have not been eradicated in India yet. Clearly, option (b) is the right answer.

46. Smallpox was declared eradicated from the world in

(a) 1975 (b) 1980
(c) 1996 (d) 2008

M.P.P.C.S. (Pre) 2017

Ans. (b)

Smallpox was an infectious disease caused by one of the two virus variants, Variola major and Variola minor. The last naturally occuring case was diagnosed in October, 1977 and the WHO certified the global eradication of the disease in 1980.

47. Consider the given statement and reason :
Assertion : Smallpox is transmitted by a virus.
Reason : The patient should rest on separate bed.

(a) Assertion and reason, both are right and the reason is based on Assertion.
(b) Assertion is true, reason is false.
(c) Assertion is false, reason is true.
(d) Assertion and reason both are false.

U.P.P.C.S. (Pre) 1990

Ans. (a)

Smallpox was caused by variola virus which was a very infectious disease. It was transmitted from person to person via infective droplets during close contact with infected symptomatic people. The patient continuously suffers from fever and red rashes that appear on the skin. The patient had to rest on separate bed. The WHO certified the global eradication of the disease in 1980.

48. Which of the following disease is not caused by a bacteria?

(a) AIDS (b) Diphtheria

(c) Cholera (d) Whooping Cough

Uttarakhand Lower Sub. (Pre) 2010

Ans. (a)

Acquired immunodeficiency syndrome (AIDS) is caused by infection with the human immunodeficiency virus (HIV). HIV infects T-helper cells, a type of T-lymphocytes which helps both B-lymphocytes (which produce antibodies) and other T-lymphocytes (which kill cells infected by viruses) to carry out their functions. Neither type of lymphocyte can therefore operate and so the body's immune system is rendered ineffective, not only against HIV but to other infections also. Due to this reason AIDS victims are frequently killed within a few years of developing the disease.

49. Consider the following statements.

AIDS is transmitted :

1. by sexual intercourse

2. by blood transfusion

3. by mosquitoes and other blood sucking insects

4. across the placenta

(a) 1, 2 and 3 are correct (b) 1, 2 and 4 are correct

(c) 1, 3 and 4 are correct (d) 1 and 3 are correct

I.A.S. (Pre) 1996

Ans. (b)

HIV/AIDS is not transmitted through sneezing, cough , kiss and the bites of mosquitoes and fleas. HIV can be transmitted from an infected person to another through :

– Blood (including menstrual blood)
– Semen
– Vaginal secretions
– Breast milk

Activities that allow HIV transmission :

– Unprotected sexual contact
– Direct blood contact, including injection drug needles, blood transfusions, accidents in healthcare settings or certain blood products
– Mother to baby (before or during birth, or through breast milk).

50. HIV does not spread AIDS through –

(a) HIV infected blood (b) Unsterilised needles

(c) Mosquito bites (d) Unprotected sex

Uttarakhand P.C.S. (Pre) 2005

Ans. (c)

See the explanation of above question.

51. Consider the following statements :

Assertion (A) : Immune system gets affected by AIDS.

Reason (R) : T-lymphocytes get completely damaged in AIDS.

Select the correct answer using the code given below :

Code :

(a) (A) is correct but (R) is not the correct explanation of (A).

(b) (A) is correct and (R) is the correct explanation of (A).

(c) (A) is correct but (R) is wrong.

(d) (A) is wrong but (R) is correct

Uttarakhand P.C.S. (Mains) 2002

Ans. (b)

T-lymphocytes are a type of lymphocyte that plays an important role in cell-mediated immunity. When AIDS disease takes place then T-4 lymphocytes (helper cells) are completely damaged and due to this immune system gets affected.

52. Consider the following conditions of a sick human body:

1. Swollen lymph nodes 2. Sweating at night

3. Loss of memory 4. Loss of weight

Which of these are symptoms of AIDS ?

(a) 1 and 2 (b) 2, 3 and 4

(c) 1, 3 and 4 (d) 1, 2, 3 and 4

I.A.S. (Pre) 2003

Ans. (d)

AIDS (Acquired immune deficiency syndrome or acquired immunodeficiency syndrome) is a syndrome caused by a virus called HIV (Human immunodeficiency virus) which destroys the disease resistance power of the human body. People infected with HIV may have the following symptoms :

– Rapid weight loss
– Recurring fever or profuse night sweats
– Extreme and unexplained tiredness
– Prolonged swelling of the lymph glands in the armpits, groin, or neck
– Diarrhoea that lasts for more than a week
– Sores of the mouth, anus, or genitals
– Pneumonia
– Red, brown, pink, or purplish blotches on or under the skin or inside the mouth, nose, or eyelids
– Memory loss, depression, and other neurologic disorders.

53. The disease caused by HIV is –

(a) Tuberculosis (b) Dysentery

(c) Cancer (d) AIDS

41st B.P.S.C. (Pre) 1996

Ans. (d)

See the explanation of above question.

54. Cause of 'AIDS' is –

(a) Bacteria (b) Fungus
(c) Virus (d) Protozoa

U.P.P.C.S (Pre) 2011
U.P.P.S.C. (GIC) 2010
Uttarakhand U.D.A./L.D.A. (Pre) 2007
U.P.P.C.S. (Pre) 1993

Ans. (c)

See the explanation of above question.

55. AIDS is the short form of which of the following diseases?

(a) Acquired Immune Deformity Syndrome
(b) Anticipated Immune Deficiency Syndrome
(c) Acquired Immune Deficiency Syndrome
(d) Abnormal Immune Deficiency Syndrome

U.P.P.C.S. (Mains) 2005

Ans. (c)

See the explanation of above question.

56. AIDS is caused by :

(a) helminth (b) bacteria
(c) fungus (d) virus
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (d)

See the explanation of above question.

57. AIDS is caused by :

(a) water (b) bacteria
(c) virus (d) fungus
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (c)

See the explanation of above question.

58. In AIDS virus, there is –

(a) DNA + Protein (b) RNA + DNA
(c) RNA + Protein (d) DNA only

U.P.P.C.S. (Mains) 2008

Ans. (c)

HIV belongs to a class of viruses known as retroviruses. Retroviruses are viruses that contain RNA (ribonucleic acid) as their genetic material. The HIV virus has a central core containing two identical RNA genomes and enzymes such as reverse transcriptase, protease and integrase.

59. Consider the following statements :

1. Adenoviruses have single-stranded DNA genomes whereas retroviruses have double-stranded DNA genomes.

2. Common cold is sometime caused by an adenovirus whereas AIDS is caused by a retrovirus.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2021

Ans. (b)

Adenoviruses are medium-sized, non-enveloped viruses with an icosahedral nucleocapsid containing a double-stranded DNA genome. In humans, more than 50 distinct adenoviral serotypes have been found to cause a wide range of illnesses, from mild respiratory infections in young children (known as the common cold) to life-threatening multi-organ disease in people with a weakened immune system.

Retrovirus is a type of virus that has RNA instead of DNA as its genetic material. It uses an enzyme called reverse transcriptase to become part of the host cells' DNA. This allows many copies of the virus to be made in the host cells. The virus that causes AIDS, the human immunodeficiency virus (HIV), is a type of retrovirus.

Hence, statement 1 is incorrect while statement 2 is correct.

60. The virus responsible for AIDS is an example of :

(a) Adenovirus (b) Mosaic virus
(c) T-even virus (d) Retrovirus

U.P. Lower Sub. (Pre) 2013

Ans. (d)

See the explanation of above question.

61. Which of the following diseases can be transmitted from one person to another through tattooing?

1. Chikungunya **2. Hepatitis B**
3. HIV-AIDS

Select the correct answer using the codes given below

(a) Only 1 (b) 2 and 3
(c) 1 and 3 (d) All of these

I.A.S. (Pre) 2013

Ans. (b)

As tattooing requires breaking the skin barrier, tattooing may carry health risks, including infection and allergic reactions. Infections that can theoretically be transmitted by the use of unsterilized tattoo equipment or contaminated ink include surface infections of the skin, hepatitis B, hepatitis C, tuberculosis and HIV. Chikungunya is a viral infection caused by the chikungunya virus which is transmitted by the bite of an infected mosquito. Hence, option (b) is the correct answer.

62. Which of the following strains of H.I.V. is dominant in India ?

(a) HIV 1A (b) HIV 1B

(c) HIV 1C (d) HIV 1D

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (c)

The human immunodeficiency virus (HIV) is the cause of one of the most destructive human pandemics in recorded history. Since it was first recognised in 1981, it has killed more than 25 million people. The three main routes of HIV transmission are 1. Contaminated blood (for example between injecting drug users) 2. Sex : vaginal, anal (and very rarely, oral) 3. From mother to child (either in pregnancy during birth or via breast milk). Molecular epidemiological analysis has identified the predominance of HIV-1 subtype C (HIV-1C) in India.

63. Who discovered the H.T.L.V. III AIDS Virus ?

(a) Robert Gallo (b) Edwerd Jenner

(c) Luck Izon Jenner (d) Robertson

U.P.P.C.S. (Pre) 1999

Ans. (a)

Robert Charles Gallo was the director of Institute of Human Virology in the University of Maryland, School of Medicine, Baltimore. He discovered Human T-cell Leukaemia Virus III. This is the factor of most dangerous disease-AIDS.

64. Most frequently used medicine for AIDS is –

(a) Zedovudine (Azidothymidine) (b) Micronazol

(c) Nanaxinel-a (d) Vinajol

Jharkhand P.C.S. (Pre) 2010

Ans. (a)

Zedovudine (Azidothymidine) is an anti-retroviral medicine which is used in HIV/AIDS.

65. Of the following, ELISA Test is performed to test –

(a) Diabetes (b) Tuberculosis

(c) AIDS (d) Typhoid

U.P.P.C.S. (Pre) 2007

U.P. Lower Sub. (Pre) 2004

Ans. (c)

ELISA is an acronym for enzyme-linked immunosorbent assay test that detects and measures antibodies in our blood. The ELISA was the first screening test widely used for HIV/AIDS because of its high sensitivity.

66. Which part of human body is infected by the virus causing Japanese encephalitis?

(a) Skin (b) Red blood cells

(c) Brain (d) Lungs

R.A.S./ R.T.S. (Pre) 2021

Ans. (c)

Japanese encephalitis (JE) is an infection of the brain caused by the Japanese Encepthalitis Virus (JEV). Japanese encephalitis virus (JEV) is a flavivirus related to dengue, yellow fever and West Nile viruses, and is spread by mosquitoes. JEV is the main cause of viral encephalitis in many countries of Asia with an estimated 68,000 clinical cases every year. While most infections result in little or no symptoms, occasional inflammation of the brain occurs. In these cases, symptoms may include headache, vomiting, fever, confusion and seizures. This occurs about 5 to 15 days after infection.

67. Japanese encephalitis is caused by :

(a) Bacteria (b) Virus

(c) Parasitic protozoan (d) Fungus

U.P.P.C.S. (Spl.) (Pre) 2008

Ans. (b)

See the explanation of above question.

68. Keeping pigs away from human settlements helps in the eradication of :

(a) Malaria (b) Japanese encephalitis

(c) Elephantiasis (d) Polio

I.A.S. (Pre) 2007

Ans. (b)

Domestic pigs and wild birds are reservoirs of the virus causing Japanese encephalitis; transmission to humans may cause severe symptoms. Among the most important factor of this disease are the Culex mosquitoes. Keeping pigs away from human settlements helps in the eradication of this disease.

69. Which of the following diseases, antibiotics cannot cure?

(a) Leprosy (b) Tetanus

(c) Measles (d) Cholera

R.A.S./R.T.S.(Pre) 2003

Ans. (c)

Measles is a highly contagious respiratory infection which is caused by a virus. It can not be cured by antibiotics whereas leprosy, tetanus and cholera are bacterial diseases and can be cured by antibiotics.

70. Measles disease is transmitted by –

(a) Virus (b) Fungus

(c) Bacteria (d) Mycoplasma

R.A.S./R.T.S. (Pre) 1997

Ans. (a)

See the explanation of above question.

71. **Which of the following is not an infectious disease?**
(a) AIDS (b) Chicken-pox
(c) Mumps (d) Herpes Simplex
Uttarakhand P.C.S. (Pre) 2010

Ans. (*)

As per the question all the given diseases are infectious diseases.

72. **'Ranikhet disease' is related to :**
(a) Chicken (b) Cow
(c) Goats (d) Horse
Uttarakhand P.C.S. (Pre) 2002

Ans. (a)

Ranikhet disease is related to chicken. It is a highly infectious disease. It is caused by a virus.

73. **Which of the following diseases of milching animals are infectious ?**
1. Foot and mouth disease 2. Anthrax
3. Black Quarter 4. Cowpox
Select the correct answer using the code given below :
(a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1 and 4 (d) 1, 2, 3 and 4
U.P.P.C.S. (Pre) 2003

Ans. (d)

Blackleg or black quarter is an infectious bacterial disease most commonly caused by Clostridium chauvoei, affecting cattle, sheep and goats. Foot and mouth disease (FMD) or hoof-and-mouth disease is an infectious and sometimes fatal viral disease that affects cloven-hoofed animals, including domestic and wild bovids. Anthrax is an acute disease caused by the bacterium Bacillus anthracis. Most forms of the disease are lethal and it affects mostly animals. Cowpox is an infectious disease caused by the cowpox virus.

74. **Foot and mouth disease in animals, a current epidemic in some parts of the world, is caused by :**
(a) Bacterium (b) Fungus
(c) Protozoan (d) Virus
I.A.S. (Pre) 2002

Ans. (d)

Foot and mouth disease (FMD) is a viral disease that affects cloven-hoofed animals, including domestic and wild bovids. FMD is caused by an Aphthovirus of the family Picornaviridae. This disease is mainly found in cattle, sheep, goat, pig etc.

75. **Foot & mouth disease is mainly found in –**
(a) Cattle & sheep (b) Cattle & pig
(c) Sheep & goat (d) All of these
M.P.P.C.S. (Pre) 1993

Ans. (d)

See the explanation of above question.

76. **The year 2011 has been marked by the livestock disease named –**
(a) Foot & mouth disease (b) Rinderpest
(c) Rabies (d) Cowpox
U.P.P.C.S.(Pre) 2012

Ans. (b)

Rinderpest also known as cattle plague was a contagious viral disease affecting cloven-hoofed animals mainly cattle and buffalo. Rinderpest is caused by a virus of the family Paramyxoviridae, genus Morbillivirus. Rinderpest was declared eradicated in 2011, making it the first animal disease to be eliminated in the history of humankind.

77. **Which of the following is a bird flu virus ?**
(a) N_5H_1 (b) NH_5
(c) HN_5 (d) H_5N_1
U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (d)

Bird flu is the prevalent name of avian influenza which is caused by H_5N_1 virus. It is a contagious disease and affects mainly birds especially hens, cock and ducks. H_5N_1 can infect human by two methods :
(1) Directly hen to man contact.
(2) Human to human transmission.
Tamiflu is an anti-viral medication that blocks the actions of influenza virus types A and B.

78. **Which of the following is responsible for 'Bird flu'?**
(a) H_5N_1 (b) H_1N_1
(c) Zika (d) Retro
U.P. R.O./A.R.O. (Pre) 2016

Ans. (a)

See the explanation of the above question.

79. **H_5N_1 virus causing global pandemic influenza is –**
(a) Goat flu (b) Bird flu
(c) Horse flu (d) Cow flu
R.A.S./R.T.S.(Pre) 2010

Ans. (b)

See the explanation of the above question.